Object Size-Aware Adaptive Scale Adjustment for RD-Curve Stabilization under Tool Interaction in VCM

Min-suk Kim, Hyo-seung Shin, Se-hee Park, Sang-woon Kwak, Soon-heung Jung and Kwang-deok Seo

***Abstract*—In the VCM-RS reference software, the Spatial Resampling scale factor is determined before RoI-based coding but applied after it. This paper analyzes the resulting tool interaction and proposes an object size-aware scale adjustment that stabilizes the RD curve, reducing BD-Rate on SFU-HW detection and in most TVD tracking configurations.**

I. Introduction

Video is increasingly consumed by machines rather than humans, motivating MPEG's Video Coding for Machines (VCM). However, CE 0 studies [1] evaluated preprocessing tools individually, leaving their interaction unexamined. This paper identifies a structural mismatch between the scale decision and the actual down-sampling as a cause of RD-curve instability and proposes an object size-aware scale adjustment.

II. Object Size-Aware Adaptive Scale Adjustment

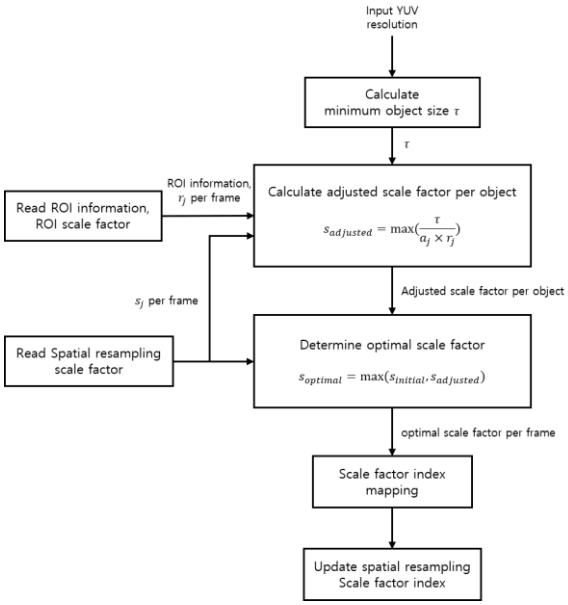


Fig. 1. Flowchart of the object size-aware scale adjustment step.

When Spatial Resampling and RoI-based coding are jointly applied, frame down-sampling and object-centric retargeting compound the reduction in object size. To prevent excessive shrinkage, we define the minimum detectable object size τ, set to 16 pixels for the detector used in this work: below this size, an object spans fewer than two cells in the finest feature map (stride 8) of a CNN-based detector and can no longer be reliably detected. Given the object size a_j (= $min(w_j, h_j)$), the RoI scaling ratio r_j [2], and the Spatial Resampling scale factor s [3], [4], the adjusted scale is computed as $s_{adjusted} = max_j(\tau/(a_j \times r_j))$, so that $a'_j = a_j \times r_j \times s$ satisfies $a'_j \geq \tau$, as shown in Fig. 1. The final scale is determined as $s_{optimal} = max(s_{initial}, s_{adjusted})$ and rounded up to the nearest value in the predefined scale factor set. This adjustment is invoked only for high-resolution inputs (frame width $W \geq W_0$), for which small-object loss is critical.

III. Experimental Results

Under the VCM Common Test Conditions [5], the proposed method reduces BD-Rate in all six SFU-HW detection configurations (up to -3.99%, LD_e2e) with BD-mAP gains; low-resolution classes are unaffected. On TVD tracking, reductions reach -7.37% (AI_e2e), though LD configurations degrade (Table I).

TABLE I. Average BD-Rate and BD-mAP/BD-MOTA of the proposed method against the conventional scale decision

	SFU-HW (detection)		TVD (tracking)	
	BD-Rate	**BD-mAP**	**BD-Rate**	**BD-MOTA**
RA_e2e	-0.65%	0.09	-6.08%	0.89
LD_e2e	-3.99%	0.72	1.39%	0.24
AI_e2e	-0.98%	0.06	-7.37%	0.34
RA_inner	-0.72%	0.06	-0.21%	0.16
LD_inner	-1.95%	0.26	15.45%	-1.26
AI_inner	-0.88%	0.05	-6.96%	1.02

Conclusion

The proposed object size-aware scale adjustment stabilizes the scale decision under tool interaction in VCM and reduces BD-Rate in most configurations on both datasets. The method has been integrated into VCM-RS from version 1.2.

Acknowledgment

This research was supported by an Institute of Information and Communications Technology Planning & Evaluation grant funded by the Korean government (MSIT) (No. 2020-0-00011, Video Coding for Machine) and by the Regional Innovation System & Education(RISE) program through the Gangwon RISE Center, funded by the Ministry of Education(MOE) and the Gangwon Sate(G.S.), Republic of Korea.(2026-RISE-10-006)

References

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